In-the-loop Machine Learning for Large-Eddy-Simulation and beyond

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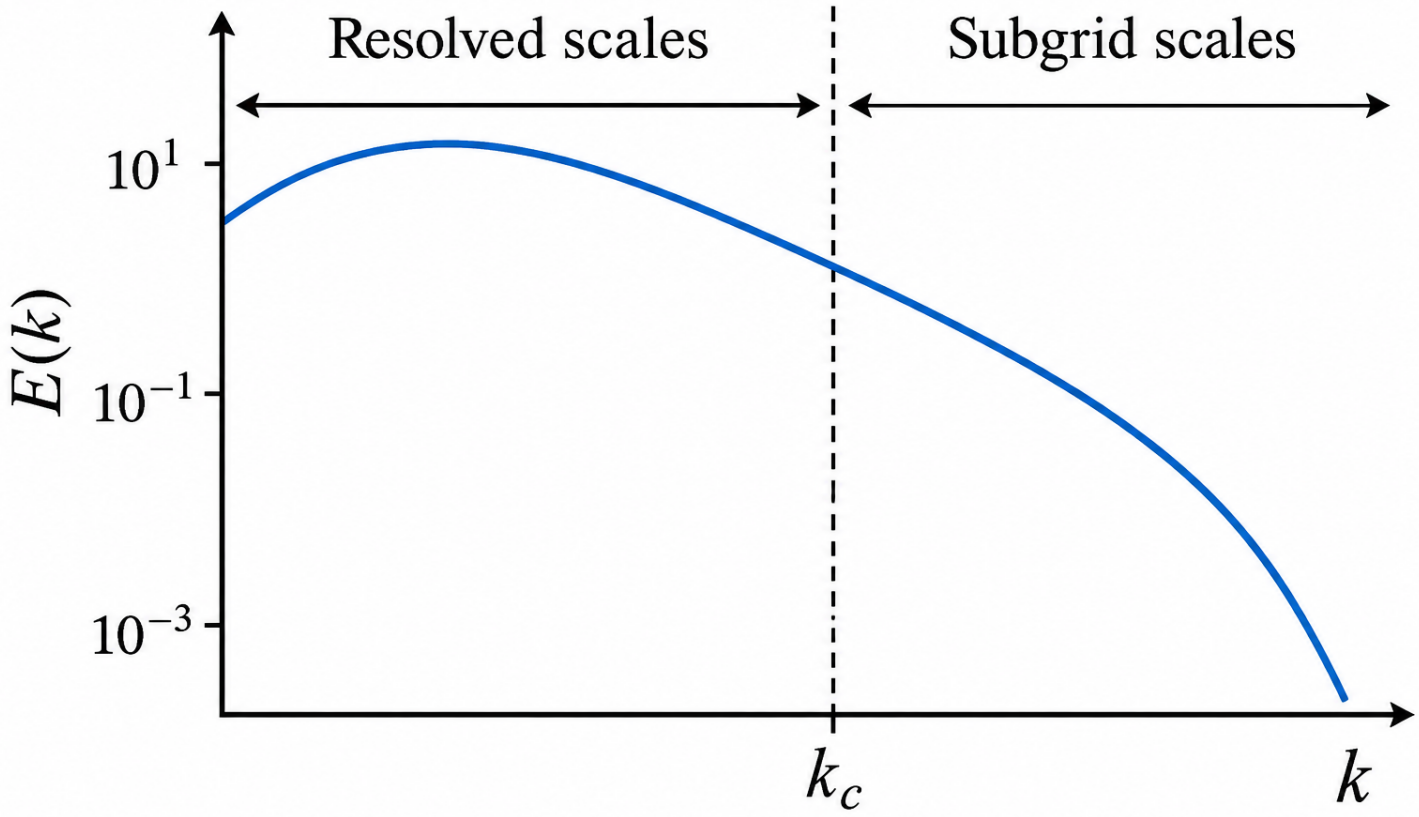
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Outline

- Overview of how we incorporate Large-Eddy-Simulations (LES) frameworks with ML and mean theories & early results.
- Simulation-in-the-loop training pipeline, challenges & potential applications.
- Team effort: Matthias Rheinhardt, Niki Loppi, Ghassem Gozaliasl, Anneliese Braunegg, Dale Weight, Maarit Korpi-Lagg

Large Eddy Simulations

LES simulations [1] try to find an effective description for the effect of the unresolved scales on the resolved ones. With ML we train on large resolution simulations to learn the effective terms needed to describe the missing scales.



[1] Meneveau and Katz, 2000

Mean field MHD

$$\frac{\partial \overline{B}}{\partial t} = \nabla \times (\overline{U} \times \overline{B} + \overline{\mathcal{E}}) + \eta \nabla^2 \overline{B} \tag{1}$$

$$\overline{\mathcal{E}} = \overline{U \times B} - \overline{U} \times \overline{B} \tag{2}$$

Learning $\overline{\mathcal{E}}$, which is a statistical description, is a job for a ML model!

We want to find a better ansatz/architecture by using our physical understanding.

Mean field MHD - fluctuation evolution

$$B = \overline{B} + b \tag{3}$$

$$\overline{\mathcal{E}} = \overline{U \times B} - \overline{U} \times \overline{B} = \overline{U \times \overline{B}} + \overline{U \times b} - \overline{U} \times \overline{B}^2 \tag{4}$$

$$\frac{\partial b}{\partial t} = \nabla \times (U \times \overline{B} + U \times b - \overline{U} \times \overline{B} - \overline{\mathcal{E}}) + \eta \nabla^2 b \tag{5}$$

Evolution equation of b is linear in $\overline{B}$ with a prescribed $\overline{U}$.

[2] To use a general averaging procedure we do not assume the Reynolds averaging rules hold.

Mean field MHD - weak field limit

Weak-field limit: no backreaction on U $\rightarrow \overline{\mathcal{E}}$ becomes linear in $\overline{B}$[3]

$$\overline{\mathcal{E}}(t,x)_i = \int_0^t \int_\infty K_{ij}(x,t,\xi,\tau)\overline{B}_j(x+\xi,t-\tau)d^3\xi d\tau \tag{6}$$

$$\begin{aligned}\overline{\mathcal{E}}(t,x)_i = \int_0^t \int_\infty & \mathcal{A}_{ij}(x,t,\xi,\tau)\overline{B}_j(x+\xi,t-\tau) \\ & + \mathcal{B}_{ijk}(x,t,\xi,\tau)\partial_k\overline{B}_j(x+\xi,t-\tau)d^3\xi d\tau\end{aligned} \tag{7}$$

[3] Rädler, 2014

Mean field MHD - scale separation

K_{ij} tends to zero over turbulent decorrelation scales and times. Assuming sufficient scale separation over these scales

$$\overline{\mathcal{E}}(t,x)_i = a_{ij}\overline{B}_j(t,x) + b_{ijk}\partial_k\overline{B}_j(t,x), \tag{8}$$

where

$$a_{ij} = \int_0^t \int_\infty \mathcal{A}_{ij}(x,t,\xi,\tau)d^3\xi d\tau \tag{9}$$

$$b_{ijk} = \int_0^t \int_\infty \mathcal{B}_{ijk}(x,t,\xi,\tau)d^3\xi d\tau \tag{10}$$

Constrained models & Comparisons

With isotropic turbulence α and β can be downgraded from tensors to scalars.

$$a_{ij} = \alpha \delta_{ij} \tag{11}$$

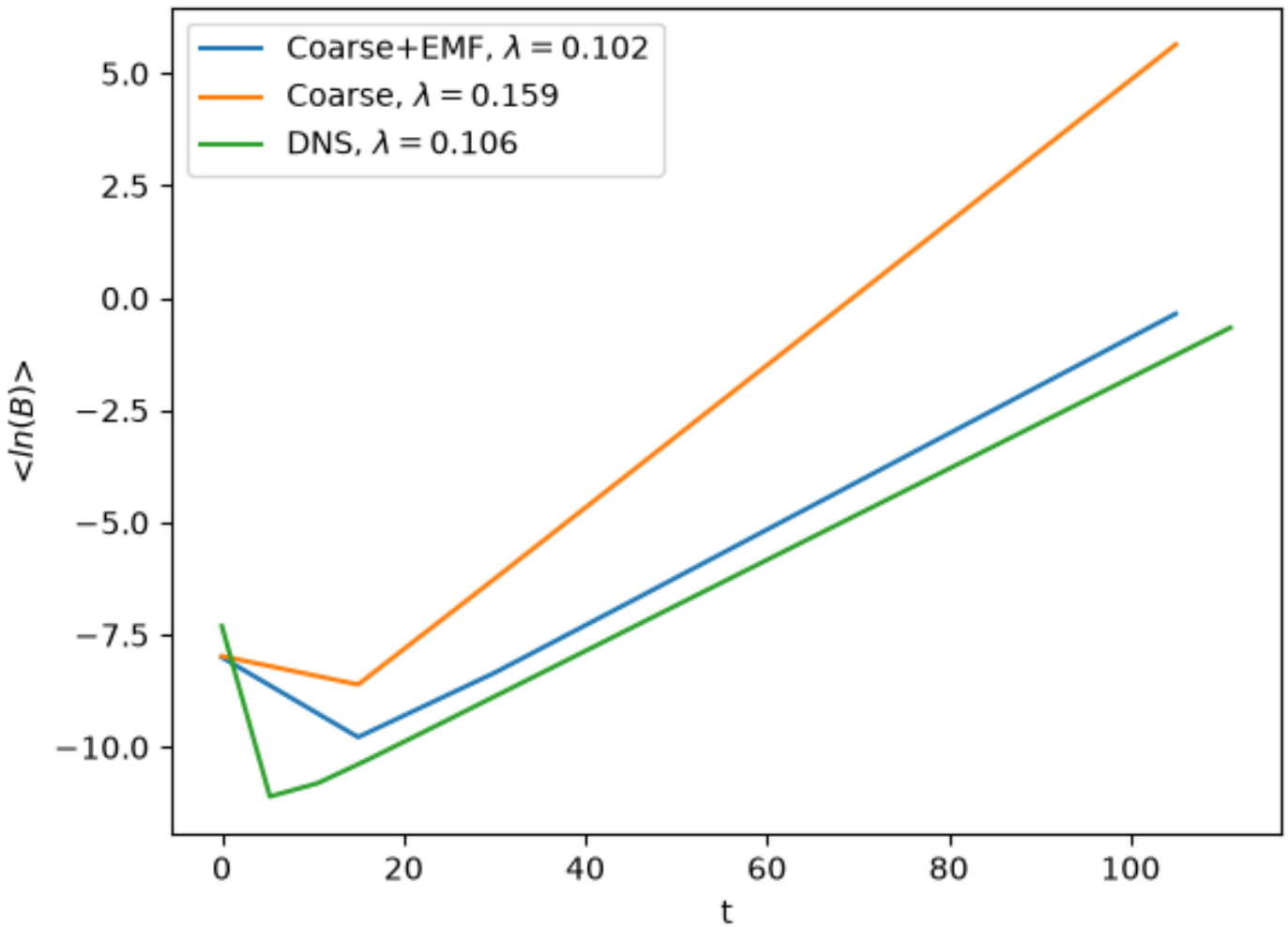
$$b_{ijk} = \beta \epsilon_{ijk} \tag{12}$$

$$\overline{\mathcal{E}} = \alpha \overline{B} - \beta \nabla \overline{B} \tag{13}$$

Good test case for ML models.
Tensors can be computed in simple situations.

Kinematic dynamo test

Trained to reproduce *DNS* solution on a 2x coarser grid.



Mean of coefficients for kinematic dynamo

$$U = (U_0 \sin(k_0 x)\cos(k_0 y), -U_0 \cos(k_0 x)\sin(k_0 y), U_z) \tag{14}$$

Symmetries can be seen from the mean tensor components

$$\begin{aligned} \langle a_{xx} \rangle &= -0.0298 \\ \langle a_{yy} \rangle &= -0.0304 \\ \langle a_{zz} \rangle &= +0.0078 \end{aligned} \tag{15}$$

$$\begin{aligned} \langle b_{xyz} \rangle &= +0.0233 \\ \langle b_{yxz} \rangle &= -0.0258 \\ \langle b_{zxy} \rangle &= +0.0220 \\ \langle b_{zyz} \rangle &= -0.0227 \end{aligned} \tag{16}$$

Needed multiple runs to constraint the coefficients enough to see the symmetries

Gleaning understanding from the models

$$\overline{\mathcal{E}} = \alpha \cdot \overline{B} - \beta \cdot (\nabla \times \overline{B}) + \gamma \times \overline{B} - \delta \times (\nabla \times \overline{B}) - \kappa \cdot (\nabla \overline{B})^{(s)} \quad (17)$$

4

- α describes how helical flows grow magnetic fields.
- β describes turbulent diffusion.
- γ describes how non-uniformity of the turbulence advects $\overline{B}$
- δ describes how rotation produces a preferred direction
- κ no clear physical interpration.

[4] Käpylä, 2025

Beyond scale separation

Two problems:

- Scale separation poorly justified for LES
- $\overline{\mathcal{E}}$ being local and instantenous in $\overline{B}$ not always well justified.

Assuming isotropy turns Eq (6) into a convolution

$$\overline{\mathcal{E}}(t,x)_i = \int_0^t \int_\infty \mathcal{A}_{ij}(\xi,\tau)\overline{B}(x+\xi,t-\tau) + \mathcal{B}_{ijk}(\xi,\tau)\partial_k \overline{B}(x+\xi,t-\tau) d^3\xi d\tau \tag{18}$$

$$\hat{\mathcal{E}}(\omega,k)_i = \hat{\mathcal{G}}_{ij}(\omega,k)\hat{B}_j(\omega,k) \tag{19}$$

Beyond scale separation-PDE

The following ansatz has been found to work well [5]

$$\hat{\mathcal{G}}(\omega, k) = \frac{a_{ij} + b_{ijk} i k_k}{1 - i\omega\tau + \ell^2 k^2}, \tag{20}$$

where ℓ and τ are proportional to decorrelation scales, most influenced by the energy containing scale k_f, following Kraichnan's random sweeping model [5] $\rightarrow$ Thus computable from $\overline{U}$.

$$(1 + \tau\frac{\partial}{\partial t} - \ell^2\nabla^2)\overline{\mathcal{E}}_i = a_{ij}\overline{B}_j + b_{ijk}\partial_k\overline{B} \tag{21}$$

The explicit temporal dependency could serve as the solution to capture transient behaviour

[5] Rheinhardt and Brandenburg, 2012

Going beyond weak field limit

Although not mathematically fully justified, make the generalization that $\tau, \ell, a_{ij}, b_{ijk}$ are all functionals of $\{u, b, \overline{U} and \overline{B}\}$[6]

Can still give physically reasonable solutions in mean-field MHD such as alpha quenching partly explaining the saturation of mean-field dynamos [7].

[6] Käpylä, 2025

[7] Käpylä, 2025

Momentum equation and Reynolds stresses

$$\frac{\partial \overline{\rho}}{\partial t} = -\nabla \cdot (\overline{M}) \tag{22}$$

$$\frac{\partial \overline{M}}{\partial t} = \nabla \cdot \left(\frac{\overline{M_i M_j}}{\overline{\rho}} \right) + \nabla \cdot \overline{P} + \nabla \cdot (2\mu \overline{S_{ij}}) - \nabla \cdot \tau_{SGS} + \nabla \cdot S_{ij_{SGS}} \tag{23}$$

where $\overline{M} = \overline{\rho U}$ and $\tau_{SGS} = \frac{\overline{M_i M_j}}{\rho} - \frac{\overline{M_i M_j}}{\overline{\rho}}$ and same for $S_{ij_{SGS}}$.

Reynolds stresses - Smagorinsky

Assume that the deviatoric part behaves as an *turbulent viscosity* [8]

$$\tau_{SGS}^{(dev)} = -2\bar{\rho}\nu_{SGS}\overline{S_{ij}} \tag{24}$$

$$\nu_{SGS} = C_s^2\Delta^2|\overline{S_{ij}}| \tag{25}$$

Can only transfer energy down to the unresolved scales!
Learn the free parameter with ML.

[8] Smagorinsky, Lilly, 1963, 1967

Residual correction

We take a baseline from the Smagorinsky model, and add a residual correction coming from the neural network.

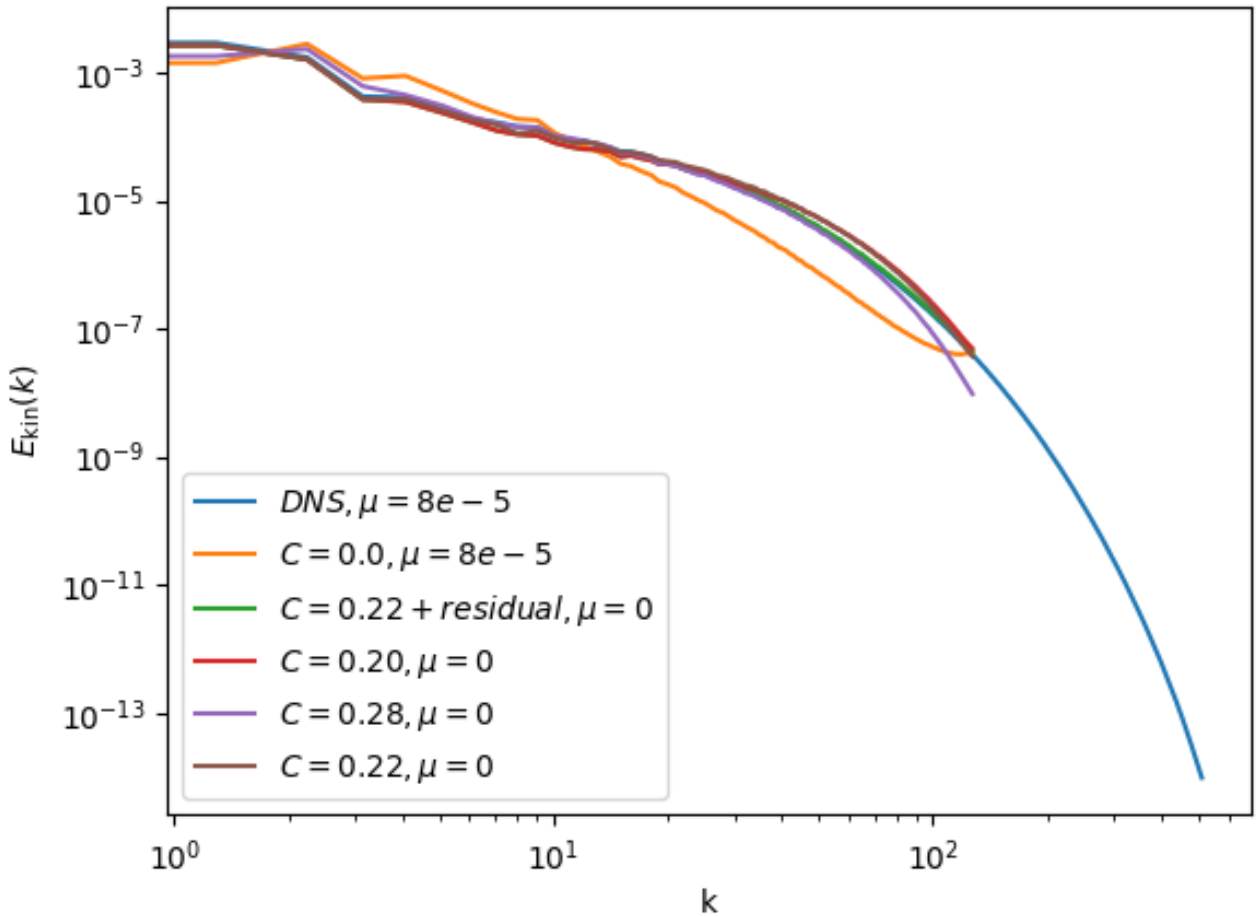
$$\tau_{SGS} = \tau_{smagorinsky} + \tau_{residual} \tag{26}$$

SGS models struggle with instabilities due to the injection of superfluous energy $\rightarrow$ constraint the magnitude of the residual to be small with a penalty term.

Comparison against literature

Train to recreate a 1024^3 result with a 256^3 grid.

Smagorinsky does most of the work, but the residual is able to add energy back to the system. Being able to rely on the energy transfer of the Smagorinsky model was very useful!



Reynolds stress - PDE

$$\frac{D\tau_{ij}}{Dt} = -\tau_{ik}\partial_k \tilde{U}_j - \tau_{jk}\partial_k \tilde{U}_i - \tau_{ij}\partial_k \tilde{U}_k - \partial_k(\overline{\rho U_i' U_j' U_k'}) + \ldots \tag{27}$$

where $\tilde{U} = \overline{M}/\overline{\rho}$ and $U' = U - \tilde{U}$ [9]

Similar equation with infinite hierarchy of average terms derivable for $\overline{\mathcal{E}}$. $\rightarrow$ Closure via ML.

[9] Germano, 1992

Connection to the Early Universe

- Relativistic hydrodynamics:
  $\partial_\eta \overline{T}^{00} = -\partial_i \overline{T}^{i0}$ [10]
  $\partial_\eta \overline{T}^{i0} = -\partial_j T^{ij}\{\overline{T}\} - \partial_j(T^{ij}\{\overline{T}\} - \overline{T}^{ij})$
- New terms to close e.g. Chiral-MHD [11]
- New degrees of freedom (scalar fields etc.) and mean theories for them to be formulated!
- Simulation-in-the-loop useful for different topics of the hackathon (UETC etc.)?

[10] Pol and Midiri, 2025
[11] Boyarsky, Fröhlich, and Ruchayskiy, 2015

ML training at exascale

Increasing our understanding of non-linear dynamics, such as turbulence, requires simulations of ever increasing fidelity.
Data analysis and data volumes have been identified as one of the prominent bottlenecks when going to exascale.

Simulation-in-the-loop training alleviates the second problem, while LES models trained using it, the first one.

Data-driven LES modeling

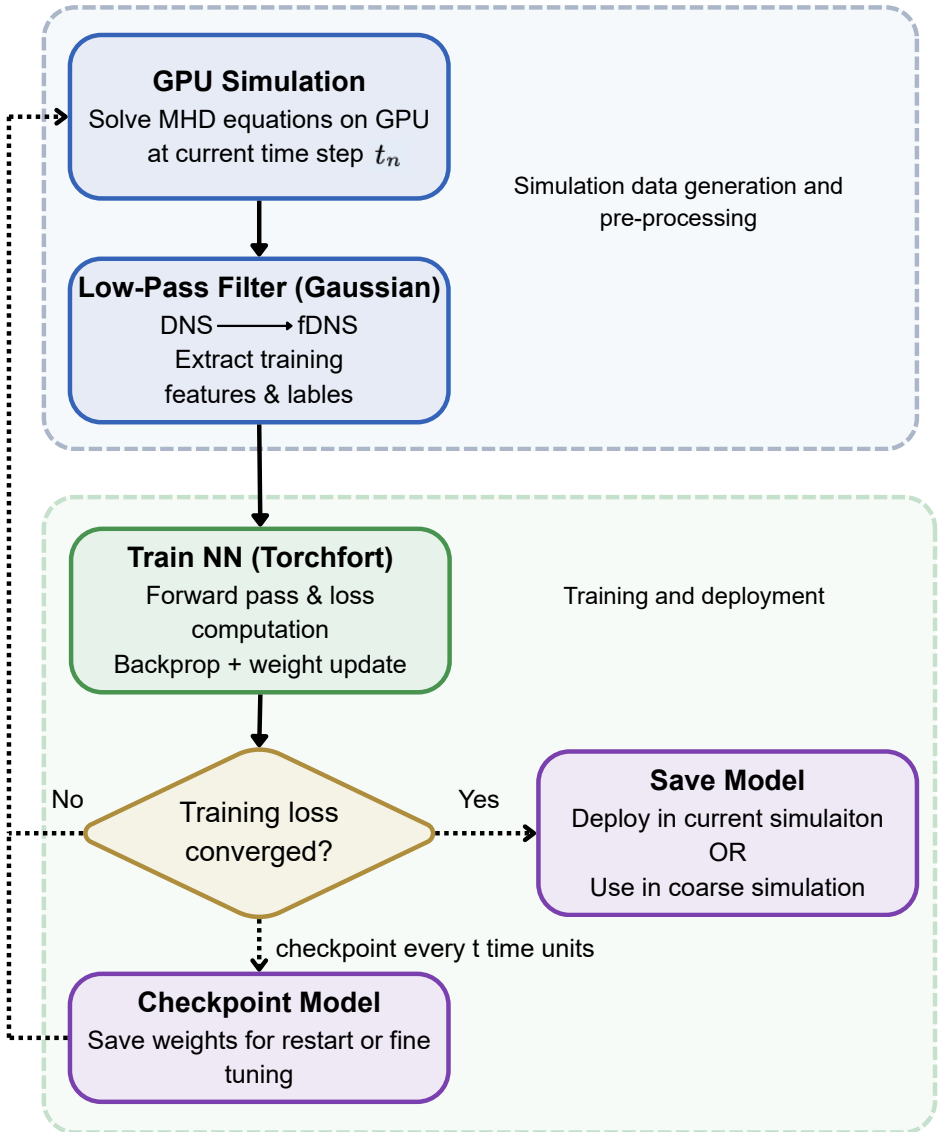
Three main types:

- using AI to learn a free parameter for closures [12]
- using AI to replace the complete closure [13]
- using AI to replace explicit solvers [14]

Exascale ML closures beyond simple parameter learning require online/in-the-loop training rather than offline learning from fixed dataset.

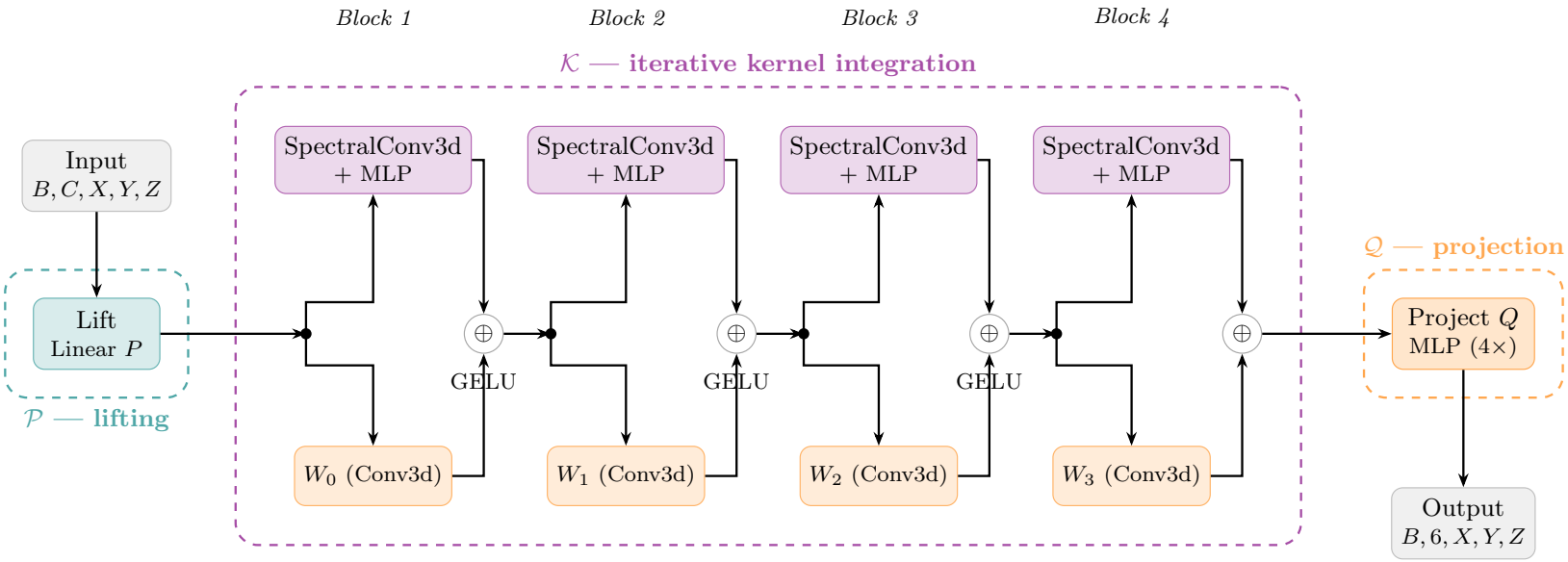
[12] Sarghini, Piomelli, and Balaras, 1999
[13] Bose and Roy, 2023
[14] Zhao et al., 2025

Simulation-in-the-loop



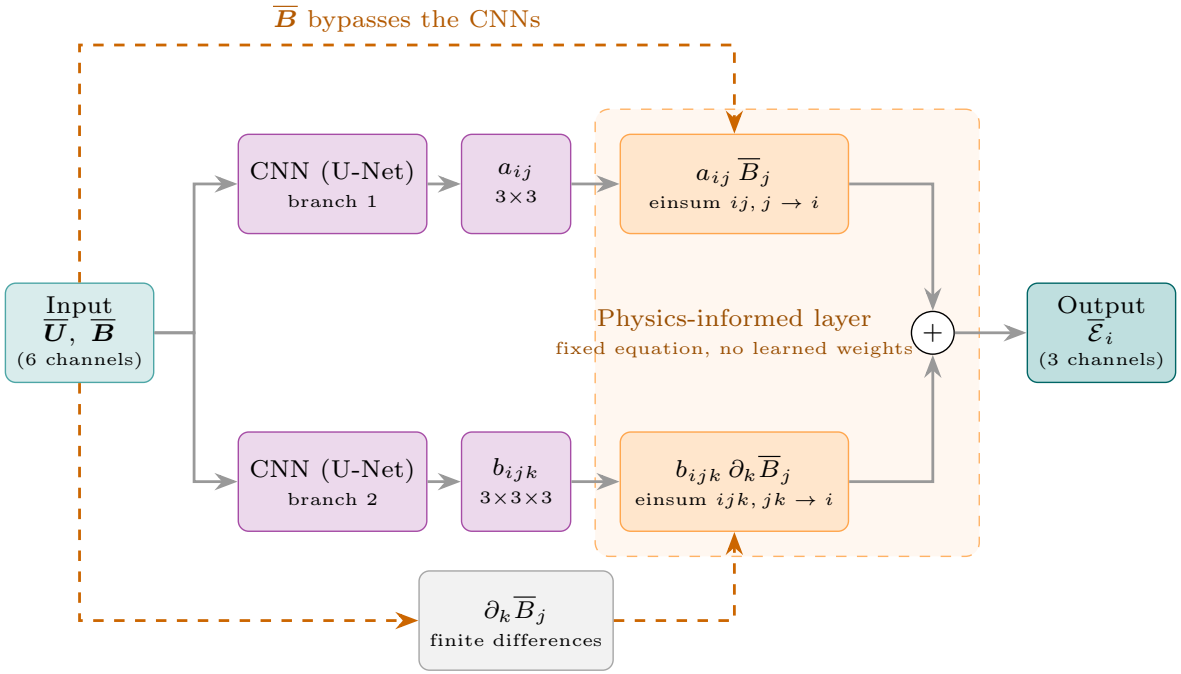
- The training and inference is coupled with the simulation's main loop, resulting is zero I/O to the disk
- Simulation through Pencil-A and training/inference through Torchfort framework
- This pipeline enables efficient scalable training and can work with any arbitrary Pytorch model and loss function

Machine Learning models: FNO / AFNO



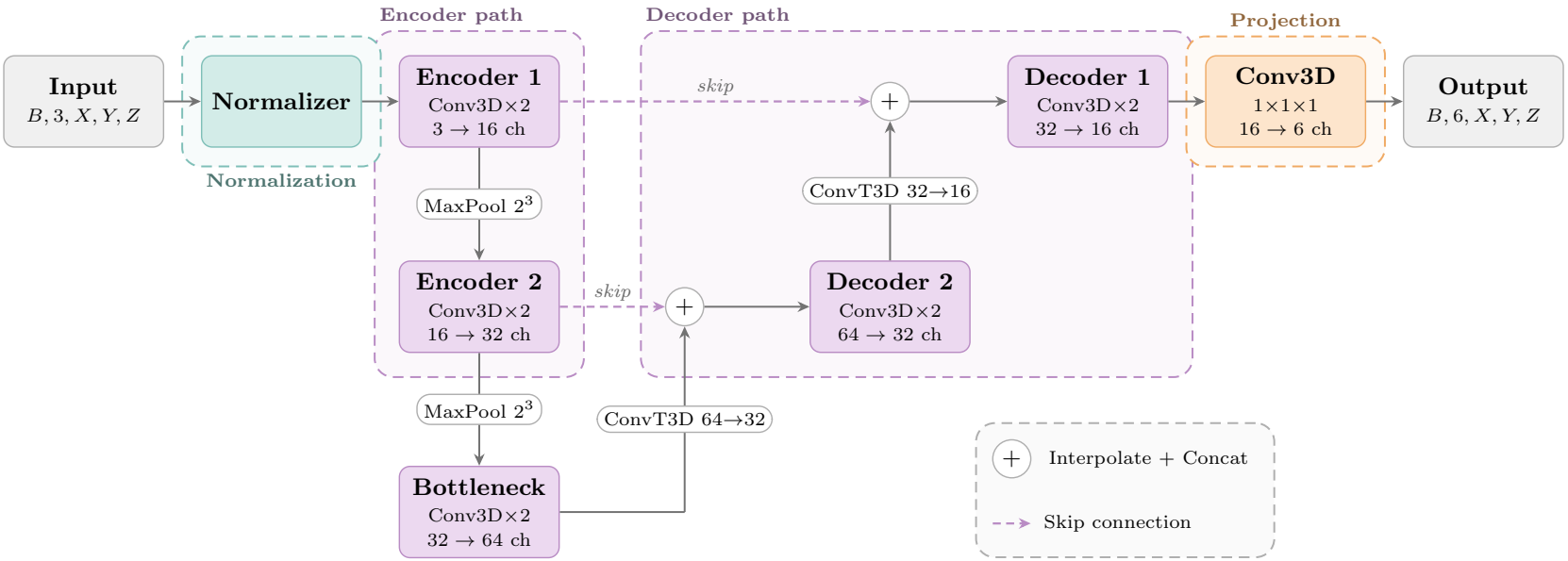
- Lift input to a latent space ($\mathcal{P}$), apply stacked Fourier layers, project back ($\mathcal{Q}$)
- Spectral convolution: global receptive field via FFT
- Pointwise W path preserves local information
- Resolution-independent; cost $\mathcal{O}(N \log N)$
- AFNO: adaptive Fourier mixing for efficiency at high resolution

Machine Learning models: Physics based Neural Network



- The CNNs learn only the coefficients a_{ij}, b_{ij} from $\overline{U}$.
- The physics informed layer applies Eq. (8), $\overline{\mathcal{E}}(t,x)_i = a_{ij}\overline{B}_j(t,x) + b_{ijk}\partial_k \overline{B}_j(t,x)$
- Enables comparison of predicted EMF and transport coefficients with reference results for model validation and interpretability.

Machine Learning models: Unet



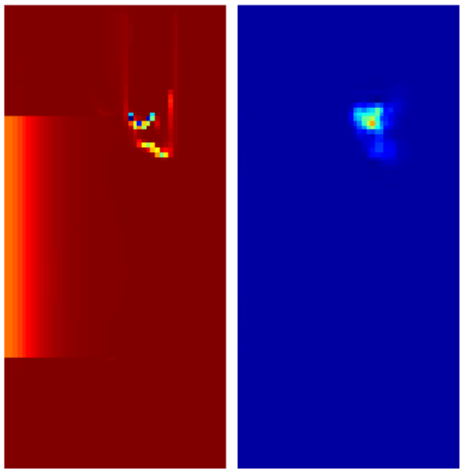
- Encoder-Decoder CNNs capture features at multiple scales
- Skip connections preserve fine-scale detail

Current challenges

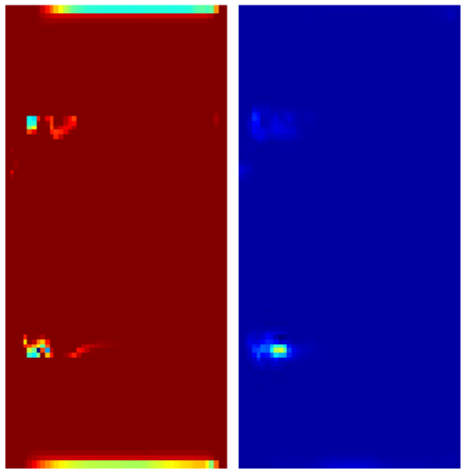
- Previous runs without any physics information results in instability when applying the model
- Capturing transient behavior is challenging
- Difficult to construct randomized mini-batch for online-training

Future applications – Numerical instablity handling

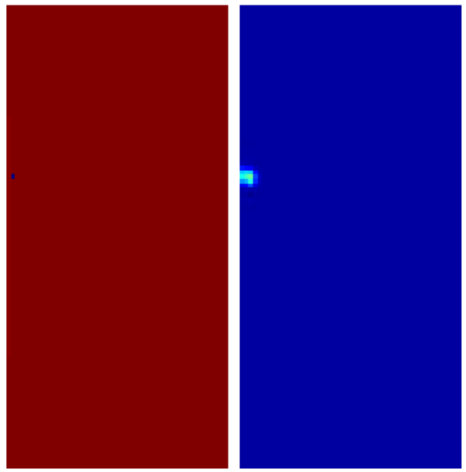
Work done by Bruno Oliveira, which are working on to incorporate to Pencil-A and to the simulation in-the-loop approach.



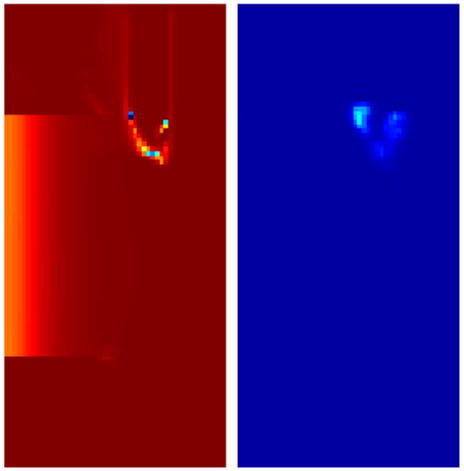
(a) Case 540



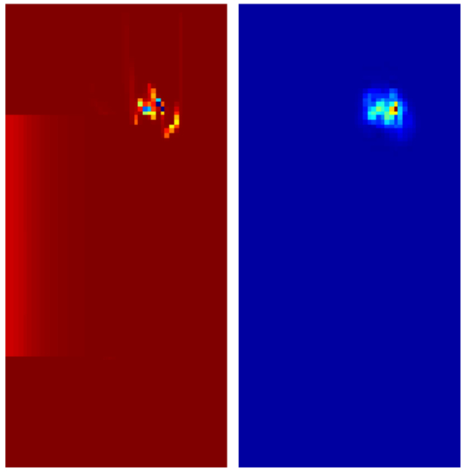
(b) Case 749



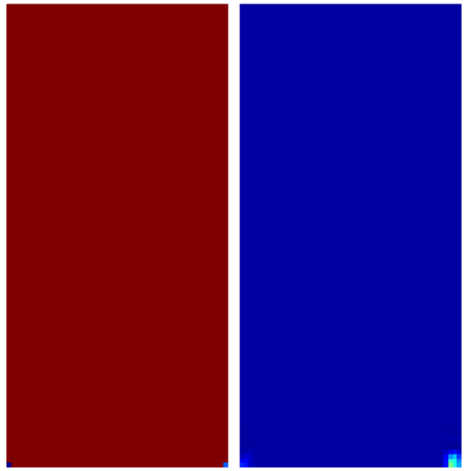
(c) Case 2393



(d) Case 2818



(e) Case 3225



(f) Case 5718

Conclusion

- With Pencil-A one can effectively train ML models alongside running simulations.
- Mean-field theories are useful in producing more physical ansatzes for ML (constraints,numerical checks,physical understanding and known properties).
- Future applications planned outside of LES
- Consider trying out the simulation-in-the-loop pipeline for your ML needs!

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